App Development Project Report

Trekking Management Application

1. Student Details

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About Me: I am **Sanu Singh**, a student of the **IIT Madras BS Degree Programme**. I developed this **Trekking Management Application** as part of the **Modern Application Development–I (MAD-1)** course. The project is built using **Flask, SQLite, SQLAlchemy, HTML, CSS, and Bootstrap**. It provides role-based access for administrators, trek staff, and users, enabling efficient management of trekking activities, bookings, and user information. Through this project, I gained practical experience in full-stack web application development, database design, authentication, and CRUD operations.

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2. Project Details

Project Title: Trekking Management Application

Problem Statement:

The objective of this project is to develop a web-based Trekking Management Application that allows administrators to manage treks and staff, enables trek staff to manage assigned treks, and allows users to browse, book, and track trekking activities.

Approach:

The application is developed using Flask with SQLAlchemy and SQLite. Role-based authentication is implemented for Admin, Trek Staff, and Users. The system supports trek management, booking, booking history, staff approval, profile management, and dashboard functionalities.

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3. AI/LLM Declaration

I used ChatGPT (GPT-5) for:

• Explaining Flask concepts
• Debugging errors
• Improving HTML/CSS layouts
• Understanding SQLAlchemy relationships
• Documentation guidance

Estimated AI usage:
15–20%

All application logic, testing, integration, and final implementation were completed by me.

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4. Technologies and Frameworks Used

Technology	Purpose
Python	Programming language used for backend development
Flask	Web framework used to build the application
SQLite	Database used to store application data
SQLAlchemy	Object Relational Mapper (ORM) used for database operations
Flask-Login	User authentication and session management
Werkzeug Security	Password hashing and verification
HTML5	Structure of web pages
CSS3	Styling and responsive design
Bootstrap 5	User interface components and responsive layouts
Jinja2	Template engine for rendering dynamic HTML pages
Visual Studio Code	Integrated Development Environment (IDE)
Git & GitHub	Version control and project repository management

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5. Database Schema / ER Diagram

Tables:

1. **Users** – Stores user information such as full name, email, phone number, password, role, account status, approval status, and account creation date.
2. **Admins** – Stores administrator credentials and account information used to manage the application
3. **Staff** – Stores trekking staff details including approval status, blacklist status, contact information, and assigned treks
4. **Treks** – Stores trek information including trek name, description, location, difficulty level, duration, slot availability, assigned staff member, trek dates, price, and current status.
5. **Bookings** – Stores trek booking details including the user, selected trek, booking date, and booking status.

Relationships:

- One-to-Many → Users → Bookings
- One-to-Many → Treks → Bookings
- One-to-Many→ Staff → Treks



6. API Resource Endpoints

Endpoint	Method	Description
`/login`	GET, POST	Authenticate Admin, Staff, and User
`/register`	GET, POST	Register a new user
`/logout`	GET	Logout the current user
`/admin/dashboard`	GET	Display Admin Dashboard
`/admin/treks`	GET	View all treks
`/admin/treks/add`	GET, POST	Add a new trek
`/admin/treks/edit/<id>`	GET, POST	Edit trek details
`/admin/treks/delete/<id>`	GET	Delete a trek
`/staff/dashboard`	GET	Display Staff Dashboard
`/staff/my_treks`	GET	View assigned treks
`/user/dashboard`	GET	Display User Dashboard
`/user/browse`	GET	Browse available treks
`/user/book/<trek_id>`	GET, POST	Book a trek
`/user/history`	GET	View trek booking history
`/user/profile`	GET, POST	View and update user profile

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7. Architecture and Features

Project Structure:

```
24f2003284.zip
│
├── app.py
├── requirements.txt
├── README.md
├── database/
│   └── db.py
├── models/
│   └── models.py
├── static/
```

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│   ├── css/
│   └── images/
├── templates/
│   ├── admin/
│   ├── staff/
│   ├── user/
│   ├── login.html
    └── register.html
```

Implemented Features

- User Registration and Login
- Admin Login
- Trek Staff Login
- Role-Based Authentication
- Trek Management (Add, Edit, Delete)
- Assign Staff to Treks
- Browse Available Treks
- Trek Booking
- Booking Cancellation
- Trek Booking History
- User Profile Management
- Staff Approval and Blacklisting
- Admin Dashboard
- Staff Dashboard
- User Dashboard
- Flash Messages for User Feedback
- Responsive User Interface using Bootstrap

Additional Features:

- Password Hashing using Werkzeug Security
- Search Functionality for Available Treks

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8. Video Presentation

Drive Link: https://drive.google.com/file/d/1uRX-dQK6kzE370Ntz1Jo0NgyMmSPL6B5/view?usp=sharing

(Accessible to all with "View" permission.)